Page 1: Deep Learning and Transformer Architectures

The Transformer is a neural network architecture based on multi-head self-attention. Unlike recurrent neural networks, Transformers process all tokens simultaneously. Self-attention computes dynamic weights between words regardless of positional distance.

Page 2: Positional Encoding and Multi-Head Attention

Positional encoding adds fixed sinusoidal signals to word embeddings because Transformers do not possess built-in recurrence or sequential order mechanisms. Multi-head attention projects queries, keys, and values into multiple subspaces. This allows the model to attend to information from different representation aspects.

Page 3: Gradient Descent and Regularization

Gradient descent optimizes neural network parameters by computing loss gradients. Adam optimizer combines momentum and adaptive learning rates for faster convergence. Dropout regularization randomly deactivates neurons during training to prevent overfitting.